Question 7

Q: A grayscale image is converted into a digital image. Describe the role of quantization in this process and its impact on image detail.

Theoretical Concept: Role of Quantization

Quantization is the process of mapping continuous intensity values (voltages from the sensor) into a discrete set of digital values (e.g., 0 to 255). It determines the **color depth** or intensity resolution of the image.

Impact: If quantization levels are too low (e.g., 2-bit or 4-bit), smooth gradients become blocky, creating "false contours" or banding artifacts. High quantization preserves fine texture details.

💻 OpenCV Python Solution

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```
import cv2  # Import OpenCV for image processing algorithms
import numpy as np                                          # Import Numpy for mat
import matplotlib.pyplot as plt                             # Import Matplotlib to

# Step 1: Create an image with complex, smooth lighting
img = np.zeros((100, 100), dtype=np.uint8)  # Create a blank black image array fill
cv2.circle(img, (50, 50), 50, 255, -1)  # Draw a filled or outlined circle on the i
img = cv2.GaussianBlur(img, (49, 49), 20) # Soft blur gradient

# Step 2: Reduce quantization from 256 levels down to just 4 levels (2-bit)
# This forces every pixel to snap to one of 4 rigid gray values
quantized_4_level = np.floor(img / 64) * 64  # Round down each pixel value to quant

# Step 3: Display the impact of poor quantization
plt.figure(figsize=(8,4))  # Create a new figure window for display
plt.subplot(121), plt.imshow(img, cmap='gray'), plt.title('256-Level Quantization')
```

```
plt.subplot(122), plt.imshow(quantized_4_level, cmap='gray'), plt.title('4-Level Qu
plt.show()  # Show the final plot window to the user
```

🎯 Expected Output

The plot compares a perfectly smooth glowing sphere with a heavily distorted sphere made of concentric rigid rings. This clearly visualizes the physical impact of low quantization limits.

🚀 How to Execute & Submit

1. Google Colab Execution:

- Open Google Colab and create a New Notebook.
- Click the 📋 **Copy** button on the code block above.
- Paste it into a Colab cell and press **Shift + Enter** to run.

2. Uploading to GitHub:

- In Colab, go to **File** > **Save a copy in GitHub**.
- Authorize your GitHub account.
- Select your Computer Vision Lab repository and click **OK** to commit the code.